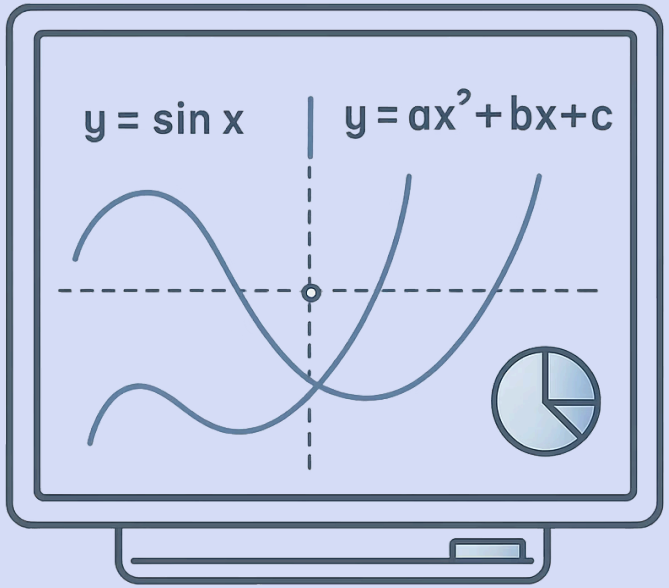
Ordinary Differential Equations

Understanding the mathematics that describes change and motion in our world



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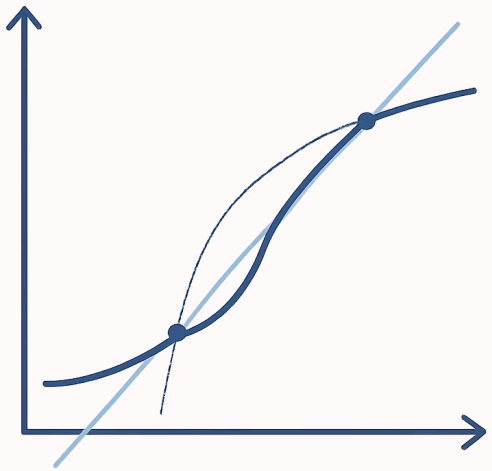
What is an ODE?

An ordinary differential equation (ODE) relates a function to its derivatives. Unlike algebraic equations that find specific values, ODEs describe how quantities change continuously.

These equations are fundamental tools for modeling dynamic systems in physics, engineering, biology, economics, and countless other fields where change is essential to understanding behavior.



Function & Derivatives

Contains unknown function and its derivatives

Single Variable

Depends on one independent variable

Modeling Tool

Describes rates of change in systems

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Classifying ODEs

Understanding the structure and properties of differential equations helps us select appropriate solution methods.

1

Order

Highest derivative present in the equation. First-order ODEs contain only first derivatives, while second-order involve second derivatives, and so forth.

2

Linearity

Linear ODEs have the unknown function and its derivatives appearing only to the first power, without products or compositions. Nonlinear equations are generally more challenging to solve.

3

Homogeneity

Homogeneous equations equal zero, while non-homogeneous equations include a forcing function. This distinction affects solution structure and methods.

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Solving First-Order ODEs

1

Separation of Variables

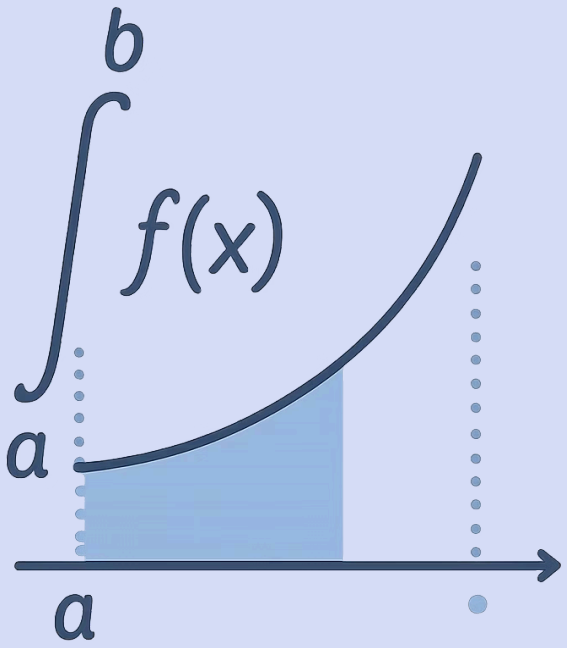
Rearrange equation so each variable appears on opposite sides, then integrate both sides independently.

2

Integrating Factors

Multiply by special function to make equation exact, enabling direct integration of the left-hand side.

These foundational techniques provide analytical solutions for many important first-order equations encountered in science and engineering applications.



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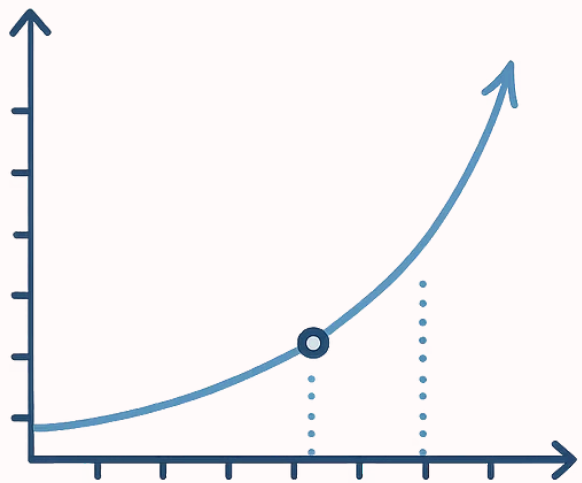
First-Order ODE Applications



Exponential Growth & Decay

Model population dynamics, radioactive decay, and compound interest using $\frac{dy}{dt} = ky$

Newton's Law of Cooling

Describe temperature change proportional to difference from ambient: $\frac{dT}{dt} = -k(T - T_{ambient})$

These real-world phenomena demonstrate how first-order ODEs capture essential relationships between quantities and their rates of change.

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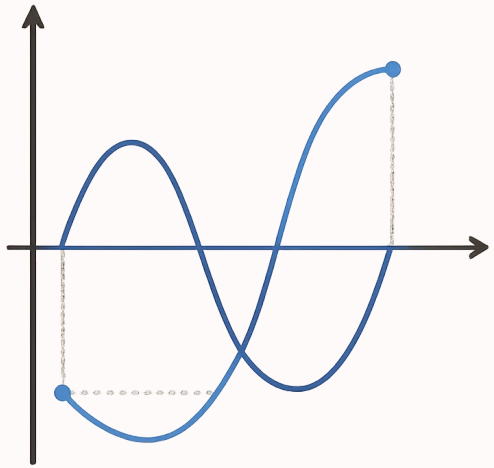
Second-Order Linear Homogeneous ODEs

Constant Coefficients

Equations of the form $ay'' + by' + cy = 0$ where a, b, and c are constants.

The solution approach involves finding the characteristic equation $ar^2 + br + c = 0$ and analyzing its roots.

- Real distinct roots: exponential solutions
- Repeated roots: polynomial-multiplied exponentials
- Complex roots: oscillatory behavior



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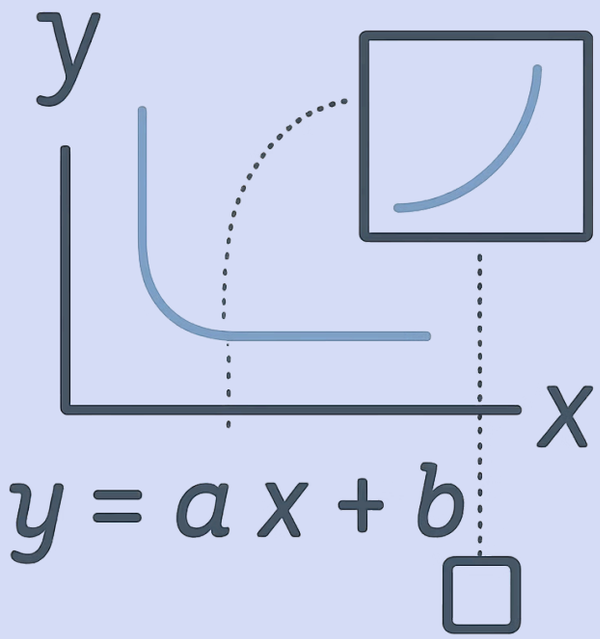

Non-Homogeneous ODEs

Method of Undetermined Coefficients

When the equation takes the form $ay'' + by' + cy = f(x)$, we find a particular solution by guessing a form similar to $f(x)$.

01

Solve Homogeneous Equation

Find general solution to corresponding homogeneous equation

02

Guess Particular Solution

Select form matching $f(x)$ structure

03

Determine Coefficients

Substitute and solve for unknown constants

04

Combine Solutions

General solution is sum of homogeneous and particular parts

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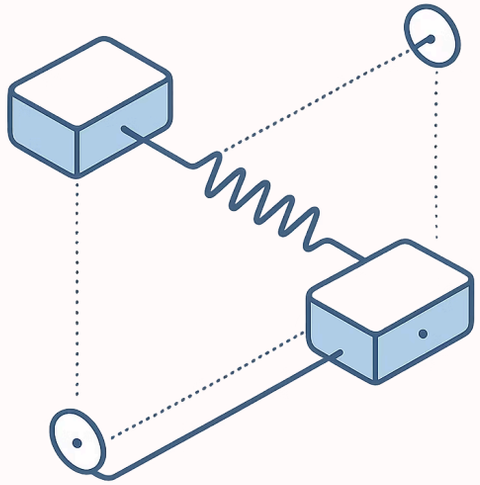
Systems of ODEs



Coupled Equations

Multiple ODEs that depend on each other describe interconnected systems. For example, predator-prey models or coupled spring-mass systems.

These systems can be written in vector form as $\frac{d\mathbf{y}}{dt} = \mathbf{f}(t, \mathbf{y})$, where $\mathbf{y}$ is a vector of dependent variables.

Interconnected

Variables influence each other's behavior

Vector Form

Compact representation using matrices

Phase Space

Solution trajectories in multi-dimensional space

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Numerical Methods for ODEs

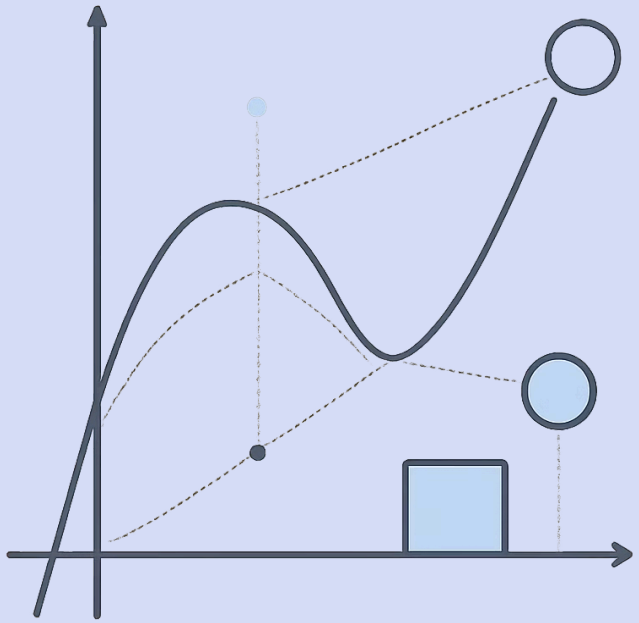
When analytical solutions are impossible or impractical, numerical methods provide approximate solutions through iterative computation.

Euler's Method

Simplest approach using tangent line approximation: $y_{n+1} = y_n + hf(t_n, y_n)$. Easy to implement but less accurate for large step sizes.

Runge-Kutta Methods

Higher-order methods like RK4 use multiple function evaluations per step for dramatically improved accuracy while maintaining stability.



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ODEs: From Theory to Impact

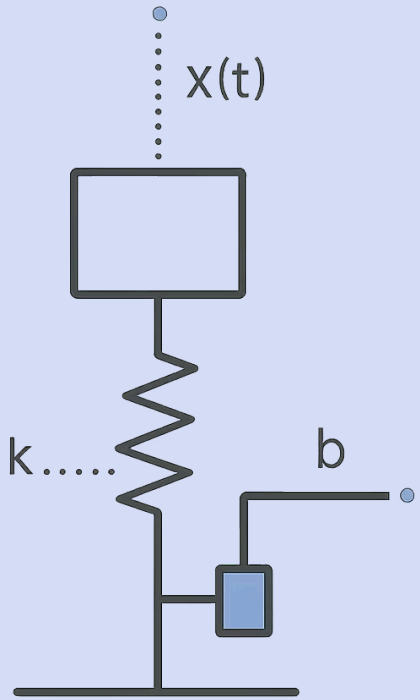
Real-World Applications

- Circuit analysis and electrical engineering
- Orbital mechanics and spacecraft trajectories
- Population dynamics and epidemiology
- Chemical reaction kinetics
- Structural vibration analysis
- Financial derivatives pricing

Advanced Topics

- Partial differential equations (PDEs)
- Nonlinear dynamics and chaos theory
- Qualitative theory and stability analysis
- Boundary value problems
- Stochastic differential equations
- Optimal control theory

Ordinary differential equations form the mathematical foundation for understanding and predicting dynamic behavior across virtually every scientific and engineering discipline.



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